TOWARDS A COMPUTER-ASSISTED EXCLUSION OF RUPERT'S PROPERTY FOR THE PARABIGYRATE RHOMBICOSIDODECAHEDRON

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Abstract. We study Rupert's property for the parabigyrate rhombicosidodecahedron, Johnson solid J73. We exhibit an order-three symmetry T of the parent rhombicosidodecahedron such that $TP \neq P$ but the projections of P and TP agree on an open set of viewing directions. The two vertex sets have forty points in common. Any box meeting this two-parameter family contains a pose with zero separation margin. We give a uniform contact-patch theorem, an exact signed-contact test, and a mixed first- and second-order tube estimate near an intersection of coincidence families. These estimates provide local exclusion criteria for a finite cover of the five-dimensional configuration space. The computation and verification of the complete cover are unfinished, so this report does not establish that J73 lacks Rupert's property.

Status of this version. The local results are proved below. Production continues, and neither a complete exclusion cover nor its exhaustive independent verification is available. Table 1 separates these obligations.

Table 1. Status at the time of this version.

Status	Scope
Proved	Translation and symmetry reduction; the open coincidence sheet; the contact-patch, signed-contact, mixed-tube, and membership results under their stated hypotheses.
Computed, not yet fully verified	Production closures and replacement trees in the frozen status snapshot; earlier retained certificates have not all passed one exhaustive independent replay.
Open	A complete finite cover; conversion and independent checking of the retained chain from every original root; the Rupert question for J73. No completion date can be given.

1. Introduction

A solid has Rupert's property if a congruent copy can pass through a straight tunnel in it with positive clearance. For convex bodies the problem can be expressed in terms of planar shadows: one asks whether an orthogonal projection of one copy can be placed strictly inside an orthogonal projection of the other. Contact alone does not give a passage. This distinction matters when a numerical search approaches scale one.

The cube is Rupert, as are the other Platonic solids. Rupert's property has also been established for many Archimedean and Johnson solids. For the history of the problem, its projection formulation, and an algorithmic approach, see Steininger and Yurkevich [2, 1]. Fredriksson used

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nonlinear optimization to find further passages [3]. Gosain and Grimmer later improved numerical passage bounds [4]; Table 4 of their cited version lists no passage for J72, J73, J74, J75, or J77.

Steininger and Yurkevich constructed the Noperthedron, a centrally symmetric convex polyhedron with 90 vertices which does not have Rupert's property [1]. Their proof combines global exclusion away from contact with local arguments near the limiting configurations. Recent preprints treat the two snub solids [5] and the rhombicosidodecahedron [6, 7]. Murphy has released Lean-based verifications for the two snub solids [8]; the kernel checks conditional theorems, while compiled programs check the certificate data. The present paper concerns J73.

We adapt Jung's computational framework to J73. The two opposite cupola gyrations preserve central symmetry and leave a proper symmetry group of order ten. An order-three rotation of the parent solid also preserves certain shadows of J73, although it does not preserve the solid. We describe this coincidence before giving the local certificates.

A complete finite exclusion cover would imply the following result.

Theorem 1.1 (Conditional completion criterion)**.** *Let P be the exact body in Section 2. Suppose the certificate consists of a finite rooted forest of closed boxes whose roots cover the reduced domain* (7)*. Every nonterminal node has finitely many closed children whose union is exactly that node's box; each replacement is bound to the identical parent region it replaces. Suppose every reachable leaf has an exclusion certificate for its entire box, using a proved local rule whose hypotheses and whole-box membership are verified for P, with no unresolved or missing nodes. Then P is not Rupert.*

Proof. By Lemma 2.1, a passage would give strict containment at a point of the reduced domain. A root contains that point. At each nonterminal node, at least one child contains it because the children cover that node. The forest is finite and acyclic, so this process terminates at a leaf containing the point, also when it lies on a shared boundary. The verified rule for that box excludes strict containment, a contradiction. □

The complete cover required by Theorem 1.1 has not yet been verified.

2. J73 and its symmetry reduction

2.1. **Exact coordinates.** Put $s = \sqrt{5}$ and $\phi = (1 + s)/2$. Let V_0 consist of the distinct signed cyclic permutations of

$$(1, 1, \phi^3), \qquad (\phi^2, \phi, 2\phi), \qquad (2 + \phi, 0, \phi^2), \tag{1}$$

with independent signs on nonzero entries. Sort these sixty points lexicographically in the real embedding of $\mathbb{Q}(s)$, retaining these indices after the construction below. Put $a = (-\phi, -1, 0)$ and $h = \max_{v \in V_0} a \cdot v$. Define

$$G = \frac{\phi}{2} I + \frac{s-1}{4} [a]_\times + \left(\frac{1}{2} - \frac{s}{5}\right) a a^{\mathsf{T}}. \tag{2}$$

The matrix G is a rotation through $\pi/5$ about the oriented axis a. Set

$$p_i = \begin{cases} G v_i, & a \cdot v_i = h, \\ G^{\mathsf{T}} v_i, & a \cdot v_i = -h, \\ v_i, & \text{otherwise}, \end{cases} \qquad V = \{p_i : 0 \le i < 60\}, \quad P = \operatorname{conv} V. \tag{3}$$

These are the edge-length-two body coordinates of J73. Only the five vertices on each extreme pentagon change; the joining decagons are permuted by the cupola rotations. Direct substitution

gives

$$V = -V, \qquad \|p_i\|^2 = R_b^2 = 11 + 4s. \tag{4}$$

The input checker verifies that the vertices are distinct, that the 120 edges have squared length four, and that the symmetries below permute V [10]. All certificate indices retain the parent ordering.

The vertices and reference rotations used in the exact contact identities have entries in $\mathbb{Q}(s)$. The alignment matrix

$$A = \begin{pmatrix} 0 & 0 & 1 \\ -1/\sqrt{\phi+2} & \phi/\sqrt{\phi+2} & 0 \\ -\phi/\sqrt{\phi+2} & -1/\sqrt{\phi+2} & 0 \end{pmatrix} \tag{5}$$

belongs to $\mathrm{SO}(3)$ and maps a to the positive vertical axis. The pipeline uses the normalized vertices Ap_i/R_b, whose coordinates require a larger algebraic field. We use P in body coordinates for the local theorems; common scaling does not change shadow containment.

2.2. **Proper symmetries and screen frames.** The rotations

$$D = \{G^{2k}, SG^{2k} : 0 \le k < 5\}, \qquad S = \mathrm{diag}(-1, -1, 1),$$

preserve V. They satisfy $(G^2)^5 = S^2 = I$ and $SG^2S = G^{-2}$, so $D \cong D_5$ has order ten. The reduction uses only the verified subgroup D and central symmetry. In aligned coordinates the subgroup is generated by a vertical fifth-turn and a horizontal half-turn [10].

For $t, p \in \mathbb{R}$ write

$$F(t,p) = \begin{pmatrix} -\sin t & \cos t & 0 \\ -\cos t \cos p & -\sin t \cos p & \sin p \\ \cos t \sin p & \sin t \sin p & \cos p \end{pmatrix} \in \mathrm{SO}(3), \tag{6}$$

and let $M(t,p)$ be its first two rows. With $R_2(\alpha)$ a positive planar rotation, a configuration is

$$x = (t_i, p_i, t_o, p_o, \alpha), \quad K_i(x) = R_2(\alpha)M(t_i, p_i)AP, \quad K_o(x) = M(t_o, p_o)AP.$$

Here p_i, p_o denote polar angles, whereas p_j denotes a body vertex.

Lemma 2.1 (Translation and symmetry reduction)**.** *The body P is Rupert if and only if $K_i(x) \subset \operatorname{int} K_o(x)$ for some x in the closed domain*

$$\mathcal{D} = ([0, 2\pi/5] \times [0, \pi/2])^2 \times [-\pi/2, \pi/2]. \tag{7}$$

Proof. Let $K = -K$ and $L = -L$ be convex planar bodies. If $K + t \subset \operatorname{int} L$, then $K - t \subset \operatorname{int} L$ as well. The midpoint of $z + t$ and $z - t$ belongs to $\operatorname{int} L$ for every $z \in K$, since the interior is convex. Thus $K \subset \operatorname{int} L$, and translations may be removed.

Represent each spatial orientation by a proper frame. A half-turn in the aligned dihedral group moves a lower-hemisphere view to the upper hemisphere. A fifth-turn puts its azimuth in $[0, 2\pi/5]$. Apply these body symmetries independently to the two copies. Two proper frames with the same third row differ by a proper screen rotation. A common screen rotation removes the outer screen angle, leaving α as their difference. Central symmetry makes this angle periodic modulo π. This gives (7). Poles and endpoints may have several representatives, which causes no problem for a closed cover. Every configuration in the stated domain is an allowed original configuration, so the converse follows. The projection formulation of Rupert's property then gives the result [1]. □

3. The hidden coincidence

3.1. **An order-three rotation.** Consider the proper rotation

$$T = \frac{1}{4}\begin{pmatrix} 1+s & s-1 & -2 \\ 1-s & -2 & -1-s \\ -2 & 1+s & 1-s \end{pmatrix}. \tag{8}$$

It satisfies $T^{\mathsf{T}}T = I$, $\det T = 1$, $T^3 = I$, and $\operatorname{tr} T = 0$. It is therefore a rotation through 120 degrees. It permutes V_0, but

$$|V \cap TV| = 40, \qquad |V \cup TV| = 80. \tag{9}$$

In particular, it is not a J73 symmetry. Some earlier records use the improper matrix $H = -T$. Since $P = -P$, we have $HP = TP$; throughout this section we use $T \in \mathrm{SO}(3)$ [11].

Let $W = V \cup TV$, and take the cyclic index list

$$(22, 28, 16, 20, 32, 33, 41, 29, 35, 37, 31, 43, 39, 27, 26, 18, 30, 24). \tag{10}$$

Writing it as $(i_0, \dots, i_{17})$, with cyclic subscripts, define, for $0 \le j < 18$, $z \in W$, and $w \in \mathbb{R}^3$,

$$N_{j,z} = (p_{i_{j+1}} - p_{i_j}) \times (z - p_{i_j}), \qquad \mathcal{C} = \{w : w \cdot N_{j,z} > 0 \text{ whenever } N_{j,z} \neq 0\}.$$

Proposition 3.1 (An open coincidence sheet)**.** *The cone $\mathcal{C}$ is nonempty and open. For every $w \in \mathcal{C}$, the orthogonal projections of P and TP onto $w^\perp$ coincide and have boundary cycle* (10)*.*

Proof. All eighteen listed vertices belong to $V \cap TV$. At

$$w_* = (-1007120, -93249, 276631)/1048576,$$

the $18 \cdot 80$ scalar products defining the cone have 1404 positive values and 36 zeros. In every zero case $N_{j,z} = 0$: these are the two endpoints of each edge. The least positive value is

$$(-1470249 + 730489\sqrt{5})/1048576 > 0.$$

The discovery program reconstructs these scalar products from the stated coordinates and boundary cycle, (1)–(10), and determines their signs exactly in $\mathbb{Q}(\sqrt{5})$. The release run returned 1,404 positive values; the remaining 36 cases have $N_{j,z} = 0$. For any w in the cone, each listed edge is a supporting edge of $\operatorname{conv} \operatorname{proj}_w W$, and every other projected point is on its strict inner side. The projected cycle is consequently a convex eighteen-gon containing $\operatorname{proj}_w W$. Its vertices belong to both shadows, which proves equality. Finally, $\mathcal{C}$ is a finite intersection of open linear half-spaces containing w_*. Its intersection with the sphere is a nonempty open set of viewing directions. □

For each $w \in \mathcal{C}$, the largest scale allowed under non-strict containment at relative rotation T is one. Scale one is attained because the shadows coincide, and no larger scale is possible because their areas are equal. This conclusion concerns only the relative poses specified above. More precisely, let

$$\lambda_T(w) = \sup\{r \ge 0 : \exists t \in w^\perp,\ r \operatorname{proj}_w(TP) + t \subseteq \operatorname{proj}_w P\}.$$

Then $\lambda_T(w) = 1$ for $w \in \mathcal{C}$; this is not a maximization over all pairs of orientations. A related distinction between contact and passage occurs in the author's earlier study of a touching arc of the snub cube [9]. Here the coincidence directions form an open subset of the sphere rather than an arc.

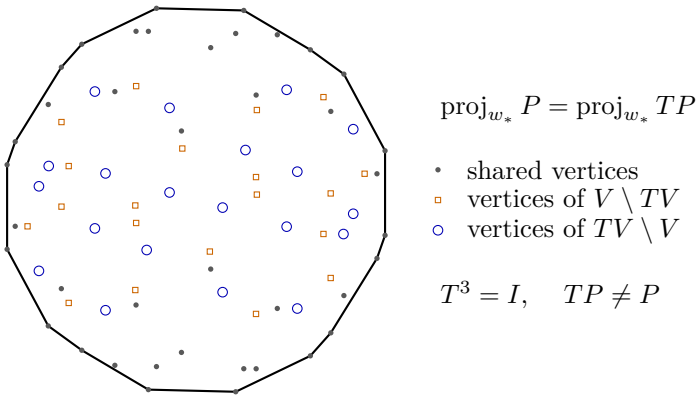


FIGURE 1. The common eighteen-edge shadow at w_*. The nonshared vertices project inside it. The equality persists on an open set of views, by Proposition 3.1. Plot coordinates are rounded for display; the proof uses the exact scalar products, not this drawing.

3.2. **Fifty rotations and four hundred labels.** The double orbit

$$\mathcal{O} = \{g_o T g_i^{\mathsf{T}} : g_o, g_i \in D\} \tag{11}$$

contains fifty distinct rotations. Each has two representatives among the one hundred ordered group pairs; none belongs to D. The exact finite enumeration and vertex counts are supplied in [11]. If $U = g_o T g_i^{\mathsf{T}}$, then

$$\mathrm{proj}_{g_o w}(UP) = \mathrm{proj}_{g_o w} P \qquad (w \in \mathcal{C}).$$

The corresponding inner view in its own body coordinates is $g_i T^{\mathsf{T}} w$. An oriented pose record also retains a view sign $\sigma \in \{-1, 1\}$ and a screen lift $L \in \{I, \mathrm{diag}(-1, -1, 1)\}$. At the anchor the triples

$$(\sigma g_o w_*,\ g_o T g_i^{\mathsf{T}},\ L)$$

give four hundred distinct labels. These labels record the viewing direction and screen lift as well as the relative rotation; labels with the same rotation may describe different view patches.

Any box meeting the coincidence sheet contains a pose with zero separation margin. Exclusion near the sheet therefore requires an estimate that permits equality on the sheet and controls nearby rotations.

At w_* the discovery data also give four common contact vertices. Set $(i_1, i_2, i_3, i_4) = (26, 31, 39, 30)$ and $q_j = p_{i_j}$ for $1 \le j \le 4$. Let $n_j \perp w_*$ be the uniquely supporting covector at q_j. For the torques $b_j = q_j \times n_j$, the exact data give weights $c_j > 0$ with

$$\sum_j c_j = 1, \qquad \sum_j c_j b_j = 0, \qquad \det(b_1, b_2, b_3) \neq 0.$$

For every nonzero infinitesimal rotation ω, some $b_j \cdot \omega$ is positive. Otherwise their positive weighted sum would force all of them to vanish, contrary to the rank condition. This is a fixed-view, first-order statement, not a uniform neighborhood certificate. The included program `code/discovery/check.py` checks these identities independently of production. Its reproduction record is given with the frozen release data.

4. GLOBAL AND LOCAL CERTIFICATES

4.1. **Support differences.** For a compact convex body K, let $h_K(n) = \max_{q \in K} n \cdot q$. For $w \neq 0$ and $n \perp w$, this is also the support value of $\mathrm{proj}_w K$. If $n \neq 0$, $n \perp w$, and an inner point

q satisfies $n \cdot q \geq h_P(n)$, then strict centered containment is impossible. Central symmetry then excludes translated strict containment by Lemma 2.1. We first give a sufficient condition based on Taylor's formula.

Lemma 4.1 (A global Taylor certificate)**.** *Let $B = \prod_{d=1}^{5}[c_d - h_d, c_d + h_d]$. For a fixed inner vertex and fixed nonzero planar covector n, put $f_j(x) = n \cdot (z_i(x) - z_j^o(x))$, where z_i, z_j^o are the inner and outer projected vertices. Suppose verified bounds $M_{de,j}$ satisfy $|\partial_d \partial_e f_j(x)| \leq M_{de,j}$ on B and, for every outer vertex j,*

$$f_j(c) - \sum_d |\partial_d f_j(c)| h_d - \frac{1}{2} \sum_{d,e} M_{de,j} h_d h_e \geq 0.$$

Then no configuration in B gives strict shadow containment.

Proof. Taylor's formula along the segment from c to x gives the displayed quantity as a lower bound for $f_j(x)$. Thus the inner point has support value at least that of every outer vertex, for the same n. □

The release uses four certificate categories. G checks a support separation on the entire original angle box. SC checks a nonnegative weighted sum of exact contact differences. $COVER$ reconstructs closed splits and coordinate covers. The *tube* rule checks a certified local region together with whole-box membership. These are the categories implemented by the included independent checker; they are described in Section 6.2. Older production records use additional labels, including S, $L2$, $L5$, balls, and collision witnesses. Their presence in a historical producer record is not acceptance by the new checker. Every retained witness must either be replayed under its original proved rule or be converted and rechecked before it can contribute to a complete independently verified cover.

4.2. **The contact-patch theorem.** The theorem allows any finite number of contacts. Our implementation often uses four, but four contacts need not suffice.

Theorem 4.2 (Uniform contact patch)**.** *Let $P = \operatorname{conv} V = -P$ be a full-dimensional polytope, with $\|q\| \leq R_b$ for $q \in V$. Let B be a closed polygon, and let $w : B \to \mathbb{R}^3 \setminus \{0\}$ be affine. Fix $U \in \mathrm{SO}(3)$ and pairs $p_j, q_j \in V$ satisfying $Up_j = q_j$. Suppose affine covectors $N_j(w) \perp w$ satisfy, on all of B,*

$$N_j(w) \neq 0, \qquad N_j(w) \cdot (q_j - v) \geq 0 \quad (v \in V), \tag{12}$$

$$\max_j (q_j \times N_j(w)) \cdot z \geq c\|z\| \quad (z \in \mathbb{R}^3), \tag{13}$$

$$\|N_j(w)\| \leq M. \tag{14}$$

Let $c, C, r > 0$ satisfy $C \geq R_b M/2$, $r \leq \pi$, and $Cr < c$. For every $w \in w(B)$ and $E \in \mathrm{SO}(3)$ with $\operatorname{ang} E \leq r$, $\operatorname{proj}_w(EUP)$ cannot be translated strictly inside $\operatorname{proj}_w P$. If $\operatorname{ang} E > 0$, a selected inner point lies strictly beyond an outer support line; at $E = I$ there is exact contact.

Proof. Write $E = \exp([\omega]_\times)$, with $\theta = \|\omega\| = \operatorname{ang} E$. The integral remainder gives

$$E - I - [\omega]_\times = \int_0^1 (1-t) \exp(t[\omega]_\times)[\omega]_\times^2 \, dt, \qquad \|E - I - [\omega]_\times\|_2 \leq \theta^2/2.$$

Consequently

$$N_j \cdot (Eq_j - q_j) = (q_j \times N_j) \cdot \omega + R_j, \qquad |R_j| \leq C\theta^2.$$

For $\theta > 0$, (13) selects a j for which this is at least $c\theta - C\theta^2 \geq \theta(c - Cr) > 0$. Equation (12) makes N_j an actual outer support covector, so this excludes containment. At $\theta = 0$, any selected shared support point excludes strict containment. Translation is removed by central symmetry as before. □

For rational polygonal patches and affine data in a specified ordered algebraic field, the hypotheses admit finite sufficient tests. For example, $N_j = d_j \times w$ makes the support slacks affine, so polygon-vertex tests suffice for the weak inequalities in (12). The separate condition $N_j \neq 0$ must hold on the entire patch; a strictly positive lower bound for $\|N_j\|^2$ is a sufficient test. If the torque hull $\operatorname{conv}\{q_j \times N_j(w)\}$ contains the radius-c ball, then (13) holds. For four torques $G_1, \dots, G_4$, verify that they form a nondegenerate tetrahedron containing that ball. For each face opposite G_i, choose its three other indices a, b, d and a consistent orientation of

$$H_i = \pm(G_b - G_a) \times (G_d - G_a), \qquad d_i = H_i \cdot G_a.$$

Throughout the patch require $H_i \neq 0$, $d_i > 0$, $H_i \cdot G_i < d_i$, and $d_i \geq c\|H_i\|$ for every i. The strict omitted-vertex inequalities exclude coplanar degeneracy. Exact Bernstein bounds on the face polynomials and squared norm inequalities give sufficient tests over the specified coefficient field. An anchor ball with an explicit uniform variation bound is another test [12].

The theorem does not assume equality of the full shadows at U. It still applies when an additional outer point becomes active, provided the weak support inequalities remain valid. The criterion requires a uniform constant $c > 0$. Moreover, a positively spanning set of torques need not contain four torques whose convex hull contains the origin in its interior; the set $\{\pm e_1, \pm e_2, \pm e_3\}$ is an example.

4.3. **Exact signed contacts.** For $u \in \mathbb{R}^3$ let

$$C(u)a = \frac{(1 - |u|^2)a + 2u(u \cdot a) + 2u \times a}{1 + |u|^2}. \tag{15}$$

This is the Cayley rotation. Its denominator is positive for every u.

Theorem 4.3 (Signed-contact certificate)**.** *Let $P = \operatorname{conv} V = -P$ be full-dimensional, and let a closed parameter cell B have exact maps $w(\xi) \neq 0$ and $u(\xi)$. Fix $U \in \mathrm{SO}(3)$. For each of finitely many candidates choose $p_j, q_j \in V$ and $N_j(\xi) \perp w(\xi)$, and put*

$$a_j = Up_j, \quad H_j = N_j \cdot q_j, \quad b_j = N_j \cdot (a_j - q_j).$$

Suppose constant weights $\lambda_j \geq 0$ have $\sum_j \lambda_j > 0$. For each positive-weight candidate suppose, on the entire closed cell,

$$H_j > 0, \qquad N_j \cdot (q_j - v) \geq 0 \quad (v \in V). \tag{16}$$

Define the exact numerators

$$F_j = (1 + |u|^2)b_j + 2(a_j \times N_j) \cdot u + 2(N_j \cdot u)(a_j \cdot u) - 2|u|^2 N_j \cdot a_j. \tag{17}$$

If $\sum_j \lambda_j F_j \geq 0$ on B, then none of its configurations allows translated strict containment of $\operatorname{proj}_w(C(u)UP)$ in $\operatorname{proj}_w P$.

Proof. Substitution of (15) gives the identity

$$F_j = (1 + |u|^2)N_j \cdot (C(u)a_j - q_j).$$

The support line is valid and its covector nonzero by (16). Strict centered containment would make every positive-weight F_j negative. Their nonnegative weighted sum, with positive total weight, would then be negative, a contradiction. Central symmetry removes translations. □

The theorem does not require $Up_j = q_j$, so the constant term b_j is retained. A vanishing weighted sum does not by itself prove equality of the shadows.

When the vertices, U, weights, covector maps, and polynomial cell maps have coefficients in $\mathbb{Q}(\sqrt{5})$, exact Bernstein coefficients give finite sufficient tests on rational boxes. Transform to $[0,1]^d$ and write a polynomial in the tensor Bernstein basis. Nonnegative coefficients imply a

nonnegative polynomial, because the basis functions are nonnegative and sum to one. Strictly positive coefficients give the strict height test. One may first prove an exact factorization $S = t^m Q$ on a cell with $t \geq 0$ and apply the test to Q. The factorization is checked as an identity, also at $t = 0$. A singleton face is certified after exact substitution and does not certify a surrounding neighborhood. These are sufficient tests; a negative Bernstein coefficient is not a counterexample to the desired inequality [13].

4.4. **Fan cells and closed rotation cones.** An affine view chart has w_k fixed and nonzero. For a proposed support covector $N = e \times w$, the height $N \cdot q$ and all slacks $N \cdot (q - v)$ are affine in the other two view coordinates. Their zero lines subdivide a view rectangle into closed convex polygons. Split by exact half-plane intersections and retain both children. Each step preserves the parent's union; a fan triangulation of each polygon then preserves its union as well. A polygon's edges and vertices remain in the closed neighboring cells. Equality of total areas alone is not a cover proof.

For a triangle with vertices v_0, v_1, v_2, the map

$$w(r,t) = v_0 + r(v_1 - v_0) + (1-r)t(v_2 - v_0), \quad 0 \leq r, t \leq 1,$$

covers the closed triangle. Support tests at its three vertices suffice for affine slacks. Rotation cubes have six closed charts:

$$u = \tau z, \quad z_k = \epsilon, \quad |z_j| \leq 1 \ (j \neq k), \quad 0 \leq \tau \leq R_c, \quad k = 1,2,3, \quad \epsilon = \pm 1. \tag{18}$$

Indeed, for $u \neq 0$ choose a coordinate attaining $|u|_\infty$, set $\tau = |u|_\infty$, and take $z = u/\tau$. Zero belongs to all six charts. Thus a signed-contact certificate on every triangle–cone product excludes the product enclosure. Here R_c is a Cayley half-width, not an angle; the angle is at most $2\arctan(\sqrt{3}R_c)$.

A product enclosure may meet a coincidence branch even when the image of the original angle box does not. In that case, an unsuccessful certificate search on the enclosure gives no conclusion about the original box. The root-zero fan study provides such an example [14]. One must refine the angle box or preserve more of the correlation in its image.

5. A thickened mixed-barrier tube

Near the intersection considered here, first-order estimates alone do not exclude all relative rotations. The following estimate combines first- and second-order terms for the specified body P, reference rotation T, and coordinates [15, 16].

Put $J_x = \mathrm{diag}(-1,1,1)$, $k = (3+\sqrt{5})/2$, $\ell = k+1$, and $H_c = 2+\sqrt{5}$. Use the body P and T above, and the chart

$$w = (-1, a, b), \qquad u = (x, y, -t), \qquad R = C(u)T.$$

For a full-dimensional centrally symmetric body define

$$\mathcal{B}(w,R) = \max_{n \perp w,\ h_P(n)=1} (h_{RP}(n) - 1). \tag{19}$$

The normalization is legitimate because the origin is interior to P.

Lemma 5.1 (Coincidence square and the second branch)**.** *For the exact input* (3)*, $TP = J_x P$. For $|a|, |b| \leq 1/100$, $\mathrm{proj}_w TP = \mathrm{proj}_w P$. Also*

$$C(0,b,-a)J_x = I - \frac{2mm^{\mathsf{T}}}{m \cdot m}, \qquad m = (1, -a, -b).$$

Consequently both $u = 0$ and $u = (0, b, -a)$ give equal shadows in this square.

Proof. Set $V_s = V \cap TV$ and $K = \operatorname{conv} V_s$. Exact vertex comparison gives $TV = J_x V$ and $J_x V_s = V_s$. Averaging a shared point with its reflection puts the polygon $K_0 = \operatorname{conv}\{(0, p_y, p_z) : p \in V_s\}$ inside K. It has twelve edges. For every nonshared vertex p, check

$$(0, p_y + ap_x, p_z + bp_x) \in K_0$$

against these edges at the four corners $(a, b) = (\pm 1/100, \pm 1/100)$. The inequalities are affine in (a, b), so these 960 finite tests cover the square. The point displayed differs from p by a multiple of w. Thus $\operatorname{proj}_w P = \operatorname{proj}_w K$. The reflection J_x changes the view chart (a, b) to $(-a, -b)$, which remains in the square. Since $J_x K = K$ and $TP = J_x P$, reflecting this equality at $(-a, -b)$ gives $\operatorname{proj}_w TP = \operatorname{proj}_w K$ as well. The input identities and edge tests are recorded in the independent review [16] and reconstructed in [15]. Finally the displayed matrix identity follows directly from (15). Its right side is reflection across $w^\perp$ and acts as the identity on that plane. Combining it with $TP = J_x P$ proves the assertion about the second branch. □

For a real number d, write $I_d = [\min(0, d), \max(0, d)]$. Let t_0 and y_0 be the nearest points to t and y in I_a and I_b. Define

$$\begin{aligned} e &= |t - t_0| + |y - y_0|, \\ Q &= (k|t_0| + \ell|y_0|)(k|a - t_0| + \ell|b - y_0|), \\ B_0 &= \ell y(a - t) + kt(b - y), \qquad D_0 = 1 + \ell at - kby, \\ X &= D_0 x - B_0, \qquad S_0 = |X| + Q. \end{aligned} \tag{20}$$

The shifted coordinate X is necessary. Replacing it with x discards terms of the same weight as the quadratic barrier.

Theorem 5.2 (Canonical mixed-barrier tube)**.** *Let $0 < \rho \le 1/9600$ and $|a|, |b|, |x|, |y|, |t| \le \rho$. Use the exact J73 input, the specified T, and the support data in Table 2. Suppose the finite support and shared-point identities described below that table are checked for this input. Then*

$$\mathcal{B}(w, C(u)T) \ge \frac{|X| + Q + e}{525} \ge 0. \tag{21}$$

Its zero set in this closed box consists precisely of $u = 0$ and $u = (0, b, -a)$. No configuration in the box allows translated strict shadow containment.

TABLE 2. Actual support witnesses for the canonical tube. Signs range independently over $\{-1, 1\}$. Here $q \in TV$ is an inner point before the Cayley perturbation, $n \perp w$, and $h = h_P(n)$.

Witness	q	n	h
Fixed	$(0, \sigma k, \ell)$	$(\sigma a + b, \sigma, 1)$	$k + \ell$
Radial	$(0, \sigma k, \ell)$	$(\sigma ka + \ell b, \sigma k, \ell)$	$k^2 + \ell^2$
Cap y	(ϵ, H_c, η)	$(a, 1, 0)$	$H_c + \lvert a\rvert$
Cap z	(ϵ, η, H_c)	$(b, 0, 1)$	$H_c + \lvert b\rvert$

The input checks verify that each listed point q belongs to $V \cap TV$ and that $h = \max_{p \in V} n \cdot p$ throughout the view square $|a|, |b| \le 1/100$. The affine dependence on a, b reduces these checks to finitely many exact comparisons in $\mathbb{Q}(\sqrt{5})$. The first two rows reduce to four corner checks. Each cap height is affine on its positive and negative half-interval; check both endpoints and zero. Lemma 5.1's finite checks are required for the claimed *equality* on the two branches, in addition to the support tests.

Proof of Theorem 5.2. For every row define the exact numerator

$$F = (n \cdot q - h) + 2(q \times n) \cdot u + 2(n \cdot u)(q \cdot u) - (n \cdot q + h)|u|^2. \tag{22}$$

It equals $(1 + |u|^2)(n \cdot C(u)q - h)$ with no omitted Taylor terms. Let M be the maximum of $F/2$ over the witnesses. We first establish

$$M \geq S_0/4 - 2016\rho e, \tag{23}$$

$$M \geq e/4 - 28\rho S_0 \qquad (e > 0). \tag{24}$$

Expanding (22), we obtain the following formulas. The fixed rows have $F/2 = A_0 \pm X$, where

$$A_0 = \ell y(b - y) + kt(a - t) - (k + \ell)x^2 + kaxy - \ell bxt.$$

The radial row with sign σ has

$$\begin{aligned} F/2 = {} & (\sigma kt + \ell y)(\sigma k(a - t) + \ell(b - y)) \\ & + (\sigma ka + \ell b)x(\sigma ky - \ell t) - (k^2 + \ell^2)x^2. \end{aligned}$$

These follow by expansion of (22). At (t_0, y_0), the maximum of the radial leading terms is Q. The diagonal part of A_0 there is nonnegative. On the full box, use $2 < k < 3$, $3 < \ell < 4$, $k + \ell < 7$, $k^2 + \ell^2 < 20$. The weak-coordinate derivatives of B_0 are at most 14ρ in absolute value, and $|B_0(t_0, y_0)| \leq Q/k \leq Q$. Since $D_0 \geq 1 - 7\rho^2 \geq 1/2$, this gives

$$|x| \leq 2S_0 + 28\rho e. \tag{25}$$

The diagonal fixed term has weak-coordinate derivatives at most 12ρ, and its remaining terms have magnitude at most $14\rho|x|$. The analogous radial bounds are 84ρ and $69\rho|x|$. Inserting (25) and weakening coefficients gives

$$M \geq |X| - 28\rho S_0 - 404\rho e, \qquad M \geq Q - 138\rho S_0 - 2016\rho e.$$

Average these and use $\rho \leq 1/552$ to obtain (23).

For the cap estimate, combine the terms in the signed maximum before bounding them. For the z cap put $z = -t$ and

$$\begin{aligned} U_c &= b(1 + x^2 - y^2 - z^2) - 2y + 2xz, \\ V_c &= 2\{x + z(y - b) + bxy\}, \\ W_c &= 1 + z^2 - x^2 - y^2 + 2by + 2bxz. \end{aligned}$$

Maximizing over its four points gives the exact numerator

$$2f_{\mathrm{cap}} = |U_c| + |V_c| + H_c W_c - (H_c + |b|)(1 + x^2 + y^2 + z^2).$$

Write $K_c = x + z(y - b)$ and $e_y = \operatorname{dist}(y, I_b)$. For $e_y > 0$, use $|V_c|/2 \geq |K_c| - \rho^2|x|$ and $|(y - b)z^2| \leq \rho(|K_c| + |x|)$ together, before discarding positive terms. On the two exterior intervals, $\epsilon = \operatorname{sgn}(b - 2y)$ satisfies $\epsilon(b - 2y) - |b| = 2e_y$. Together with

$$U_c = (b - 2y)(1 + x^2 + y^2 + z^2) + 2\{yx^2 + (y - b)(y^2 + z^2) + xz\},$$

this gives

$$f_{\mathrm{cap}} \geq (1 - 12\rho)e_y - 14\rho|x|.$$

Here the weak-coordinate loss is bounded by $(10\rho + 2\rho^2)e_y \leq 12\rho e_y$ and the x loss by $(7\rho + 7\rho^2)|x| \leq 14\rho|x|$. The y cap has the same estimate with $e_t = \operatorname{dist}(t, I_a)$, by the corresponding proper coordinate rotation of these formulas; this uses the listed cap points, not a new symmetry of P. If $e > 0$, choose the cap with excess at least $e/2$. Equation (25) yields

$$M \geq (1/2 - 6\rho - 392\rho^2)e - 28\rho S_0 \geq e/4 - 28\rho S_0,$$

which is (24). It is not asserted for $e = 0$.

For $e > 0$ average (23) and (24):

$$M \geq (1/8 - 14\rho)S_0 + (1/8 - 1008\rho)e \geq (S_0 + e)/50.$$

For $e = 0$, (23) is stronger. The last inequalities and all earlier constant bounds hold at $\rho = 1/9600$ and hence below it. Every chosen height is positive and $h(1+|u|^2) < 20(1+3\rho^2) < 21$. Some numerator F is at least $(S_0 + e)/25$; dividing that row by its own height and Cayley denominator proves (21).

If $S_0 + e = 0$, then $(t, y) = (t_0, y_0)$, and $Q = 0$ forces $(t, y) = (0, 0)$ or (a, b). In both cases $B_0 = 0$, so $X = 0$ and $D_0 > 0$ give $x = 0$. Conversely Lemma 5.1 proves equality of the whole shadows on these two branches, hence $\mathcal{B} = 0$. Everywhere else the lower bound is positive. A nonnegative support barrier excludes strict containment; central symmetry removes translations. □

The weights of a, b, t, y are one and that of the strong coordinate x is two. The clamp in (20) is part of the definition. For example, if $b = 0$ then $Q = k^2|t_0||a - t_0|$; the replacement $t_0 = t$ requires $t \in I_a$.

The theorem treats a degeneracy of the first-order support estimates within the T-coincidence set. Indeed, Lemma 5.1 gives equal shadows on the larger view square. Other centers require an exact symmetry transport or new witnesses. The bound $1/9600$ is the largest half-width allowed by the displayed conservative constants, not an optimal geometric tube width.

6. Connecting certificates to the original cover

6.1. **Whole-box membership.** Set $O = F(t_o, p_o)A$ and $I = \mathrm{diag}(R_2(\alpha), 1)F(t_i, p_i)A$. Fix $g_o, g_i \in D$, a screen lift $L \in \{I, \mathrm{diag}(-1, -1, 1)\}$, and a local reference $T_0 \in \mathrm{SO}(3)$. The canonical relative rotation and view are

$$R' = g_o^{\mathsf{T}}(LO)^{\mathsf{T}} I g_i, \quad E = R'T_0^{\mathsf{T}}, \qquad w_{\mathrm{raw}} = g_o^{\mathsf{T}} O^{\mathsf{T}} e_3. \tag{26}$$

If $1 + \mathrm{tr}\, E > 0$, its inverse Cayley coordinates are

$$u = \frac{(E_{32} - E_{23}, E_{13} - E_{31}, E_{21} - E_{12})}{1 + \mathrm{tr}\, E}. \tag{27}$$

A nonzero fixed view component gives the affine chart $w = (w_{0,k}/w_{\mathrm{raw},k})w_{\mathrm{raw}}$. We require $w_{\mathrm{raw},k} \neq 0$ and $1 + \mathrm{tr}\, E > 0$ throughout the box, with certified bounds separated from zero. Both screen lifts may be singular at the same pose; if either denominator test fails, the box requires another certified chart or remains unresolved [17].

With $U = g_o T_0 g_i^{\mathsf{T}}$ and $E_o = (LO)^{\mathsf{T}} I U^{\mathsf{T}}$, the viewing direction in the inner copy's body coordinates is

$$I^{\mathsf{T}} e_3 = U^{\mathsf{T}} E_o^{\mathsf{T}} O^{\mathsf{T}} e_3.$$

It reduces to $U^{\mathsf{T}} O^{\mathsf{T}} e_3$ if and only if $E_o^{\mathsf{T}} v = v$, where $v = O^{\mathsf{T}} e_3$; equivalently, E_o fixes v. In particular, $E_o = I$ suffices. In general, the full expression is required.

Lemma 6.1 (Affine Frobenius membership bound)**.** *Let $B = \prod_{d=1}^{5}[c_d - h_d, c_d + h_d]$ have exact endpoints, and fix $L \in \{I, \mathrm{diag}(-1, -1, 1)\}$ and $U \in \mathrm{SO}(3)$. Put $f(x) = I(x) - LO(x)U$, $J_d = \partial_d f(c)$, $a_* = h_1 + h_2 + h_5$, and $b_* = h_3 + h_4$. Then on B,*

$$\frac{\|f(x)\|_F}{\sqrt{8}} \leq \max_{\epsilon \in \{-1,1\}^5} \frac{\|f(c) + \sum_d \epsilon_d h_d J_d\|_F}{\sqrt{8}} + \frac{a_*^2 + b_*^2}{4} =: Y. \tag{28}$$

If verified bounds give $Y < 1$ and $2Y/\sqrt{1 - Y^2} \leq r$, every relative rotation $(LO(x))^{\mathsf{T}} I(x) U^{\mathsf{T}}$ has angle at most r. Together with a verified view-chart inclusion in a patch, these tests give whole-box membership in that patch's angle tube.

Proof. Along a segment from c to x, each inner Euler factor has unit angular speed. A mixed second derivative of their product has Frobenius norm at most $\sqrt{2}$: a rotation generator has rank two, norm one, and Frobenius norm $\sqrt{2}$. The inner second derivative is therefore bounded by $\sqrt{2}a_*^2$. The outer product gives $\sqrt{2}b_*^2$. The integral remainder in f is at most $(\sqrt{2}/2)(a_*^2+b_*^2)$. The norm of the affine part is convex in each coordinate, so its maximum on the box is bounded by its 32 corner values. Division by $\sqrt{8}$ gives (28). Orthogonal invariance identifies the left side with $\sin(\theta/2)$ for the relative angle $0 \le \theta \le \pi$. Finally $\theta = 2\arcsin(\sin(\theta/2))$ and $\arcsin z \le z/\sqrt{1-z^2}$ for $0 \le z < 1$. □

For signed or mixed-tube cells, (26)–(27) are evaluated by outward interval arithmetic on the whole original box. The image must lie in the certified cell or in the union of a verified closed cell cover; inclusion in its bounding box alone is insufficient. The new coordinates do not change the original region's identity. For a whole-box test it suffices that

$$\exists \text{ label } \lambda \quad \forall x \in B : \Phi_\lambda(x) \in C_\lambda.$$

Different children may use different labels. For the canonical examples checked in [16], two labels mapped the entire box into the certified cube; the images under the other 398 labels were disjoint from the cube.

6.2. **Cover and certificate interface.** For a complete proof, the verifier must check the exact input, the local exclusion theorem, and whole-box membership for every terminal box. It must also verify a finite, acyclic rooted forest covering all original roots: each nonterminal node is the union of its finitely many closed children, each replacement is bound to the identical parent region, and every reachable leaf is certified, with no unresolved or missing nodes. These are the hypotheses of Theorem 1.1. The release checker accepts four certificate types:

Rule	Obligation
G	Recompute one support separation with derivative and remainder bounds on the complete angle box.
SC	Reconstruct the support polynomials, nonnegative weights with positive total weight, and an exact nonnegative sum.
$COVER$	Reconstruct every closed child and verify its union with the other children is the identical parent; check all view triangles, rotation cones, and original-angle image conditions.
tube	Recompute a local contact-facet or canonical mixed-region certificate and show the complete angle box lies in it, with the stated proper symmetry and screen lift.

The uniform contact patch supplies the contact-facet region theorem; Theorem 5.2 supplies the canonical mixed theorem. Both use the single `j45-region-v1` region envelope. A region theorem alone does not close a box: its coordinate connection must also be checked. A box certificate alone does not verify the original grid: every preceding replacement link must also be checked.

A complete verification must reconstruct the body, reduced domain, original cover, and every replacement, then check each terminal exclusion and its whole-box connection. The full-chain replay is unfinished; the local checks reported here do not satisfy Theorem 1.1.

Legacy balls, tapers, and collision witnesses have not all been converted to this interface. Their earlier mathematical dependencies remain necessary wherever those witnesses are retained. In particular, a taper parameter endpoint requires coverage, and a collision proof may require a nearest-reference restriction and a Hoffman bound. The release does not replace these obligations with a saved producer success flag.

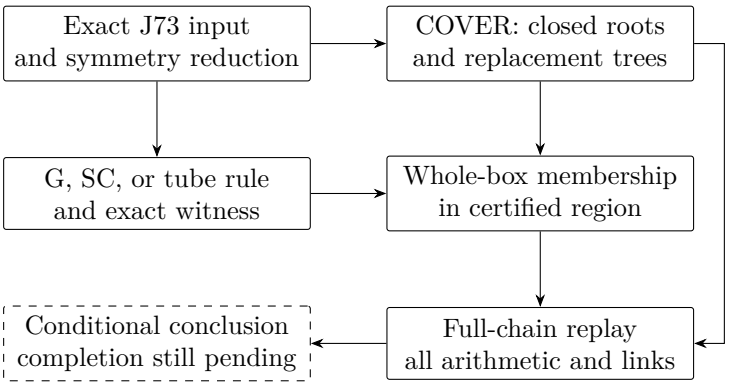


FIGURE 2. The proof obligations and the four certificate categories. Local acceptance includes whole-box membership. Full-chain replay and the dashed conditional conclusion remain unfinished.

7. The computation

7.1. **Initial grid and replacement trees.** The initial rational overcover uses a proved rational upper bound $q_+ > \pi$, obtained with a Machin formula and alternating-series bounds. The rectangle

$$[0, 2q_+/5] \times [0, q_+/2] \times [0, 2q_+/5] \times [0, q_+/2] \times [-q_+/2, q_+/2]$$

is split into $12 \times 17 \times 12 \times 17 \times 30 = 1{,}248{,}480$ rational atoms. The root order is part of the input declaration. Stored binary64 boxes round outward from these atoms, so they form an overcover and may overlap in positive volume. Within a replacement tree, the exact cuts reconstruct the parent union [10, 21].

The first traversal finished, but left unresolved roots. Its reported 48,895,305 accepted leaves are a historical producer count, not a final arithmetic verification result [18]. Later queues replace only the unresolved leaves. Subdivision followed by certification of some children can increase the number of unresolved leaves while reducing the region they cover. We therefore distinguish original roots, replacement inputs, terminal leaves, and pending tasks. None of their count ratios measures the remaining geometric volume or difficulty.

The production search uses finite-time budgets, checkpointed trees, and successive follow-up stages. The engine and storage format have changed during the run. Checksums are computed from the decompressed certificate data, so they do not depend on the compression format [22]. Exceptions, time limits, unvisited stack entries, and inconclusive signs all remain unresolved regions.

8. Interim results

8.1. **Frozen production status.** The release snapshot was captured on 27 October 2026, 10:40:50.233026–10:41:21.245658 KST, without interrupting production. Each root's committed logical proof was read through the compact-storage API with read-only database connections. This is a rolling capture of individual committed versions, not an atomic filesystem snapshot. The coordinator counts and state were unchanged across capture, although its progress heartbeat changed. The snapshot's per-root times and logical SHA-256 digests are retained in `data/status-snapshot/`.

Here "closed" means closed in the production records. A tube-assisted job contains at least one tube operation and may also contain G, SC, or $COVER$ steps; these counts have not been verified by a full-chain replay.

TABLE 3. Production counters in the frozen release snapshot. These are not exhaustive independent verification results.

Quantity	Count
Staged input jobs	2,057
Closed staged jobs	453
Open staged jobs	1,604
Pending boxes in captured open proofs	397,906
Closed receipts without a tube operation	409
Closed receipts containing a tube operation	44
Closed before coordinator stage 2	392
Additional closures by capture	61
Coordinator stage-2 visits recorded	136 / 1,665

The recorded proof budget and the time at which a job closed are different stage notions. Grouping the captured proofs by their recorded budget gives 431 closed and 1,523 open at stage 1, and 22 closed and 81 open at stage 2. The stage-1 closed receipts comprise 392 without and 39 with a tube operation; stage 2 comprises 17 without and 5 with a tube operation. Tube-only visits can retain an earlier budget, so these figures are not a second chronology of stage closures. Likewise the coordinator's 136 visits are not 136 closures.

The exact summary bytes are identified by SHA-256

`145d4f6cd74f64b01a0166dbc548a0121f92c87c45a29fbacfd814116c10dbf3`.

The manifest also binds the open-job list, compressed per-job status, public capture records, source evidence, logical proof-version hashes, and selected-certificate provenance. The raw controller record is not distributed; the summary retains its original capture digest as provenance. Production continues after this capture; later counters are not substituted into this version.

8.2. **From the original grid to the staged jobs.** The original grid contains 1,248,480 angle roots. Its first traversal reported 1,247,167 closed roots and 1,313 roots with 67,011 unresolved descendant leaves. These are different units. A subsequent traversal processed 32,420 leaf tasks and left 34,591 unprocessed. Subdivision of the processed tasks produced 1,133,950 unresolved leaves, so the switched queue had $1,133,950 + 34,591 = 1,168,541$ inputs. Its later unresolved-output counts were

$$\begin{aligned} 1,168,541 &\longrightarrow 244,200 \longrightarrow 36,895 \longrightarrow 32,332 \\ &\longrightarrow 22,053 \longrightarrow 4,759 \longrightarrow 4,105 \longrightarrow 3,524. \end{aligned}$$

These transitions replace unresolved output populations rather than counts of original roots. In the transition from 32,332 inputs, 19,666 input jobs closed, leaving 12,666 input jobs whose unresolved descendants numbered 22,053. Thus subtracting closed inputs does not give the output-leaf count.

The later static ledgers partition those 3,524 residual jobs as follows:

$$\begin{aligned} 3,524 - 15 - 25 - 1,401 &= 2,083, \\ 2,083 - 1 &= 2,082, \\ 2,082 - 4 - 21 &= 2,057. \end{aligned}$$

The first three reductions remove 15, 25, and 1,401 jobs. A separately recorded closure of job 2411 leaves 2,082; removing another 4 and 21 jobs gives the staged population of 2,057. The accompanying accounting table identifies these disjoint sets. This checks the ledger, not the underlying certificates.

8.3. **Tube closures and the remaining frontier.** The earlier tube study used a different frozen rolling snapshot, captured on 26 October 2026, 10:25:18.766258–10:25:49.262151 KST. Among its 1,662 open staged jobs and 358,906 pending boxes, the verified region union covered 7,450 boxes and every pending box of 39 jobs. The independent review and subsequent checker record checked the 1,365 tube applications in those complete frontiers [20]. This is a local frontier result; it does not replay all earlier source, G, or SC witnesses for 39 original-grid roots. These historical counts are not added to the current 44 tube-assisted producer receipts.

A tube certificate combines a local exclusion theorem with a whole-box membership test. The contact-facet theorem permits any exact proper reference rotation for which the stated support, positive torque-span, and remainder conditions can be verified. Symmetry transport must use a proper symmetry of J73 and must transport the viewing region as well. The canonical mixed theorem covers its specified intersection; translating that model to an arbitrary apparent coincidence is not justified.

Round 8 examined the remaining frontier by family, depth, and exact scale-one witnesses. Numerical proximity to a catalogue rotation is a candidate label, not a proof of an exact coincidence in the box. Conversely, a box labelled "none" has not been proved disjoint from all coincidence families. The frozen diagnostic tables and their times are given below; they are distinct from the current production counters.

The round 8 diagnostic is the rolling snapshot captured on 27 October 2026, 01:16:44–01:17:26 KST. It contained 1,615 open staged roots and 370,364 pending boxes. These historical counts are not the current counters in Table 3. Its source records and SHA-256 digests are retained with the release provenance.

TABLE 4. Historical round 8 frontier: numerical family labels and rigorous scale-one witnesses.

Label	Open boxes	Roots met	$\lambda = 1$ boxes
T	55,032	558	3,596
D5	27,666	250	0
H1	38,082	243	0
CT	50,593	203	0
TC	11,295	137	0
H4	2,555	54	0
identity	5,452	267	5,452
other	16,142	178	0
none	163,547	935	2,122
Total boxes	370,364		11,170

The family label uses the closer of the two screen lifts and the nearest catalogue rotation at the box midpoint, with a cutoff of 0.01 radians; exact identity intersections take precedence. The labels $T, D5, H1, CT, TC, H4$ are names in that fixed catalogue, "other" collects its remaining families, and "none" means outside the cutoff. The box classes are disjoint, but one root can meet several classes, so the root column cannot be added. These midpoint labels do not certify incidence. In particular, 2,122 boxes labelled "none" have rigorous T witnesses.

The rigorous witnesses establish scale one in 11,170 boxes of 577 roots: 5,452 identity boxes and 5,718 T boxes. The identity and T root counts, 267 and 316, overlap. Identity witnesses use exact rational intersections. The nonidentity witnesses use a rational outer angle, an outward-rounded inverse-frame enclosure inside the original box, and exact symmetry or rigorous common-hull support inequalities giving equality of the complete shadows. A catalogue

TABLE 5. Depth in the same historical round 8 snapshot.

Depth	Open boxes	Roots met	$\lambda = 1$ boxes
0–9	289,143	1,348	9,043
10–19	79,239	580	2,092
20–29	1,650	29	35
30	332	23	0

match alone is insufficient. The other 359,194 boxes are unclassified with respect to existence of a scale-one point; no rigorous negative census was completed. Each reported positive witness passed its construction or recovery check. The later separate cross-check repeated every identity witness and a sample of 120 T witnesses, rather than a second exhaustive replay of the latter. This release includes the historical summaries and source hashes, but omits the 3,304,005-byte witness file. Its format is not supported by the exclusion checker, and the historical equality witnesses were not replayed for this release.

All pending boxes of the 23 roots in the depth-30 row were at that depth. Their small widths do not close their roots. Reapplying an unchanged depth-limited search does not by itself resolve them. The diagnostic studies found some additional finite tube covers in private copies, but those covers were not production commits or independent acceptances of the complete original-grid chain, and are not added to the production closure counter here.

The remaining roots present several different obstacles. Some boxes meet an exact equality family, so a strictly positive support gap is impossible there. Signed-contact certificates allow a nonnegative sum, but the required exact cancellations and nonnegative polynomial decomposition still have to be found. A polynomial with an interior zero set need not admit a finite proof by nonnegative Bernstein coefficients on axis-aligned boxes in the existing coordinates. Some product enclosures also include unreachable configurations; retaining the original-angle correlations or constructively subdividing the cover can then matter more than increasing precision.

Other roots have large shallow boxes, varying support sets, or weak first-order torque span. A contact region can be valid yet fail to contain the entire original angle box. Near an intersection of equality branches, a uniform positive first-order bound can fail and a mixed estimate is needed. Some production leaves have reached the present subdivision depth; more elapsed search time alone does not create a new proof for such a leaf. None of these observations proves that a passage exists or that no certificate can be found. No completion date can be given.

9. Verification and the trust base

9.1. **Checks required for the final claim.** The checker must reconstruct V, its ordering, central symmetry, the group permutations, and the initial rational cover. It must then follow every retained old leaf and every replacement, rebuild every split, check closed boundaries, and recompute each accepted inequality and membership test. A hash identifies the data to check; it does not prove the inequality. Saved margins and PASS strings are not accepted as mathematical evidence.

Earlier audits have different scopes. The historical source inventory, recorded on 27 October 2026, reports two early samples of 40 and 60 G witnesses. Later traversals replayed their own certificate populations exhaustively, but did not certify the complete original-to-final chain [18]. The public accounting table in `data/historical-accounting.md` identifies the replacement populations. Exhaustive verification of a later traversal does not verify certificates retained from earlier stages. Atlas checks likewise establish their local regions and recorded applications,

not coverage of all unresolved roots. The final replay must include both the retained first-pass witnesses and every adopted follow-up witness.

The release replay checks three local certificates (a G leaf, an SC-based closed fan cover, and a *tube* application) and all 2,660 regions. It does not check the complete history of any original-grid root. The final verification must include every original root and adopted replacement, with no missing or unresolved region.

9.2. **Arithmetic and independence.** Exact contact identities use rational arithmetic and ordered $\mathbb{Q}(\sqrt{5})$ comparisons. New certificate inputs must use integers, canonical rational strings, or exact coefficient pairs; Boolean and floating-point inputs must be rejected. Legacy binary64 endpoints must be interpreted as the exact rationals represented by their stored bits, retaining the whole declared box.

Radicals, trigonometric functions, and coordinate maps require outward enclosures. Existing paths use Fraction-based rational bounds and Taylor remainders, or python-flint with FLINT/Arb. A working precision such as 320 bits is not itself a proof of a sign. An interval crossing zero does not prove nonnegativity; an exact zero identity does. Inconclusive comparisons must be rejected or kept unresolved.

The ordinary verifier is implemented separately from the producer [21], but the singular and newer local checkers share some geometry and arithmetic modules. The trust base includes the stated mathematical reductions and exclusion theorems, their checker implementations, parsing and cover reconstruction, integer and interval arithmetic, the interpreter and libraries, and execution on the recorded hardware. Formal proof-assistant verification of the complete J73 chain has not been supplied.

Candidate hulls, numerical optimization, linear programs, scheduling, and language-model output need not be trusted if all their witnesses are rechecked by the stated exact predicates. The discovery checker uses Python assertions. Run all checks without `-O` or `PYTHONOPTIMIZE`; `self_test.py` rejects optimized execution.

The included checker source files, their hashes, and the measured release checks are listed in the repository. Appendix A separates this local test scope from the unfinished full-chain verification.

10. Limitations

The global exclusion remains unfinished. The frontier obstacles and the scope of the local estimates are described in Section 8. Neither the contact-patch nor the signed-contact theorem has been shown to replace every legacy witness in the retained chain. A complete result still requires the dependencies and full-chain checks of Section 9.

Private historical source records are identified as not distributed in `paper/SOURCES.md`. No conclusion about J74 or J75 is drawn from this work.

11. Data and code availability

The accompanying repository contains the certificates and omissions listed in `README.md`. Its self-test checks the included local evidence, file hashes, and PDF page count. The software archive is [AUTHOR-CONFIRM: version-specific DOI]; the repository is `https://github.com/ysmin1575/j73-rupert`. The preprint DOI is [AUTHOR-CONFIRM: paper DOI reserved for the separate preprint].

The program and exact data for Section 3 are included in `code/discovery/`. The program reconstructs the vertices, rotations, sign comparisons, orbit, pose labels, and four-contact identities without importing production modules.

A future complete archive must additionally bind every original root, every adopted replacement, all retained old witnesses, and the complete verification result. The present sample or staged population does not supply that archive. The source-to-claim map for the mathematical text is `paper/SOURCES.md`.

The producer derives from Gihyo Jung's MIT-licensed project [6], pinned in the input record at `2b45714fc243c320bd9c6bae00257835755af040`. Its attribution and full MIT notice are retained in the accompanying `NOTICE`. Code is released under the MIT license. The paper text and data are released under CC BY 4.0; this does not relicense upstream code.

12. AI-use statement

Claude (Anthropic) and OpenAI Codex assisted with literature checks, mathematical derivations, code, and drafting. Mathematical acceptance requires the stated exact or outward-rounded checks rather than a language-model response. The author directed the project and takes responsibility for the submitted text. Verification of the complete certificate chain remains unfinished.

Appendix A. Release scope and outstanding verification

The release runtime uses G, SC, $COVER$, and *tube*. The exact parser, arithmetic, geometry, polynomial evidence, coordinate images, local rules, and closed-cover reconstruction belong to its trust base. The exporter, search heuristics, and stored producer verdicts do not. These boundaries are documented with the checker sources and digest manifest. The discovery program is independent of that runtime and of the producer.

Conversion of all legacy production witnesses remains in progress. The final version will give a migration record for every replaced leaf, including its unchanged parent region and whole-box membership. Until that replay succeeds, the original witness and its original proof dependencies cannot be discarded. A partial local receipt cannot be promoted to a complete-root or complete-grid verdict.

Only the software version DOI, software concept DOI, and paper DOI remain open for confirmation at upload. The software concept DOI is recorded separately in the repository README.

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[10] J73 release documentation, *Exact input and D5 reduction*, Section 2 and Section 7 of this report; accompanying `code/checker/geometry.py` and `data/status-snapshot/release-input-binding.json`. The complete original-root cover is not replayed by this release.
[11] S. Yoo, exact discovery data and independent check, incorporated in Section 3; accompanying `code/discovery/check.py` and `code/discovery/evidence.json`. The discovery note is not issued separately with this release.
[12] J73 release documentation, *Uniform contact-patch theorem*, Theorem 4.2 of this report; contact-facet checking in the accompanying `code/checker/tube.py`.
[13] J73 release documentation, *Signed-contact theorem and exact polynomial certificates*, Theorem 4.3 of this report; accompanying `code/checker/local.py` and `code/checker/polynomial.py`.
[14] J73 release documentation, *Closed fan polygons and rotation cones*, Section 4.4 of this report; accompanying `code/checker/cover.py`.
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[16] J73 historical verification records, *Mixed-barrier estimates and whole-box containment*, private records, not distributed; provenance inventory dated 27 October 2026. Source IDs `src-cbdc118ba294b922` and `src-917e39759c5a7705`; their full SHA-256 digests and byte sizes are recorded in the accompanying `paper/SOURCES.md`. The corrected arguments are in Sections 5 and 6 of this report.
[17] J73 release documentation, *Whole-box membership and coordinate connection*, Section 6 of this report; accompanying `code/checker/image.py`, `code/checker/tube.py` and `code/checker/cover.py`.
[18] J73 historical source inventory, *Proof map, certificate inventory, and trust base*, private records, not distributed; provenance inventory dated 27 October 2026. Source IDs `src-fdc77b2b3dd84f4c`, `src-14c86cbed8f0f001` and `src-ca6e021cfc12af91`; their full SHA-256 digests and byte sizes are recorded in `paper/SOURCES.md` and `data/source-evidence-manifest.json`. These historical records are not final production statistics.
[19] J73 release documentation, *Retained legacy proof obligations*, Sections 6.2 and 9 of this report; accompanying `PROOF_MAP.md`. The historical balls, tapers and collision arguments remain dependencies where retained witnesses use them; their complete payloads and original-to-final replay are not supplied by this release.
[20] J73 historical diagnostic summary, *Tube-frontier coverage for the snapshot of 26 October 2026 and independent verification*, accompanying `data/historical-diagnosis/README.md` and its provenance in `data/status-snapshot/historical-diagnosis-manifest.json`. The counts concern the captured pending frontiers, not an end-to-end acceptance of original-grid roots.
[21] J73 release documentation, *Independent J73 checker, version 2*, accompanying `code/checker/`. The release README and self-test record state its tested population; no complete-grid acceptance is asserted.
[22] J73 historical accounting and frozen status, *Production counters, root partitions, and provenance*, accompanying `data/status-snapshot/`, its SHA-256 manifest, and `data/historical-accounting.md`. The source-to-claim map is `paper/SOURCES.md`. These are production counters and ledger checks, not exhaustive verification results.

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